KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4071 | 4 Credits: 4 | Program Core | Source-grounded

Process Heat Transfer

 To impart the knowledge on different modes of heat transfer, construction and

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4071 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4071.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Chemical Engineering
Course code	4071
Course title	Process Heat Transfer
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L: 3 T: 1 P: 0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart the knowledge on different modes of heat transfer, construction and
- working of different heat exchanging equipment in industries.

Course outcomes

CO	Official outcome	Study level
CO1	12 Applying conduction and convection. Summarise the heat transfer in fluids and radiation	See official syllabus
CO2	17 Applying heat transfer.	See official syllabus
CO3	Summarise heat exchangers. 16 Applying	See official syllabus
CO4	Summarise evaporators. 15 Applying	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	convection.	4	As specified
Module 2	Summarise the heat transfer in fluids and radiation heat transfer.	4	As specified
Module 3	Summarise heat exchangers.	4	As specified
Module 4	Summarise evaporators.	4	As specified

MODULE 1

convection.

This module is presented from the official syllabus scope for Process Heat Transfer. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: convection.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Outline the basic modes of heat transfer. 1 Understanding Interpret the mechanism of heat transfer through

Outline the basic modes of heat transfer. 1 Understanding Interpret the mechanism of heat transfer through is an official topic in Module 1 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the basic modes of heat transfer. 1 Understanding Interpret the mechanism of heat transfer through using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the basic modes of heat transfer. 1 understanding interpret the mechanism of heat transfer through should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the basic modes of heat transfer. 1 Understanding Interpret the mechanism of heat transfer through means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Outline the basic modes of heat transfer. 1 Understanding Interpret the mechanism of heat transfer through □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

6 Applying conduction. Interpret the mechanism of heat transfer through

6 Applying conduction. Interpret the mechanism of heat transfer through is an official topic in Module 1 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying conduction. Interpret the mechanism of heat transfer through using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying conduction. interpret the mechanism of heat transfer through should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying conduction. Interpret the mechanism of heat transfer through means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 6 Applying conduction. Interpret the mechanism of heat transfer through □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Applying convection

3 Applying convection is an official topic in Module 1 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying convection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying convection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying convection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 3 Applying convection [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Interpret overall heat transfer coefficient. 2 Applying Heat transfer – Modes of heat transfer, definitions. Heat transfer by conduction in solids – Steady state and unsteady state flow – definition - Units of heat flow - Fourier's law of conduction – Rate equation for heat flow – Steady state heat flow - Conduction through single wall – Derivation of equation and simple problems – Thermal conductivity – Units - Steady state conduction through composite wall in series - Derivation of equation and problems - Steady state conduction through cylindrical wall and spherical wall derivation – Problems - Economic thickness of insulation – Commonly used insulating materials. Theory of convection – Film concept of heat transfer temperature gradient in forced convection – Derivation of overall heat transfer coefficient from individual heat transfer coefficient – Simple problems. Compute overall heat transfer coefficient – Simple problems.

Interpret overall heat transfer coefficient. 2 Applying Heat transfer – Modes of heat transfer, definitions. Heat transfer by conduction in solids – Steady state and unsteady state flow – definition - Units of heat flow - Fourier's law of conduction – Rate equation for heat flow – Steady state heat flow - Conduction through single wall – Derivation of equation and simple problems – Thermal conductivity – Units - Steady state conduction through composite wall in series - Derivation of equation and problems - Steady state conduction through cylindrical wall and spherical wall derivation – Problems - Economic thickness of insulation – Commonly used insulating materials. Theory of convection – Film concept of heat transfer temperature gradient in forced convection – Derivation of overall heat transfer coefficient from individual heat transfer coefficient – Simple problems. Compute overall heat transfer coefficient – Simple problems. is an official topic in Module 1 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interpret overall heat transfer coefficient. 2 Applying Heat transfer – Modes of heat transfer, definitions. Heat transfer by conduction in solids – Steady state and unsteady state flow – definition - Units of heat flow - Fourier's law of conduction – Rate equation for heat flow – Steady state heat flow -

Conduction through single wall – Derivation of equation and simple problems – Thermal conductivity – Units - Steady state conduction through composite wall in series - Derivation of equation and problems - Steady state conduction through cylindrical wall and spherical wall derivation – Problems - Economic thickness of insulation – Commonly used insulating materials. Theory of convection – Film concept of heat transfer temperature gradient in forced convection – Derivation of overall heat transfer coefficient from individual heat transfer coefficient – Simple problems. Compute overall heat transfer coefficient – Simple problems. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interpret overall heat transfer coefficient. 2 applying heat transfer – modes of heat transfer, definitions. heat transfer by conduction in solids – steady state and unsteady state flow – definition - units of heat flow - fourier’s law of conduction – rate equation for heat flow – steady state heat flow - conduction through single wall – derivation of equation and simple problems – thermal conductivity – units - steady state conduction through composite wall in series - derivation of equation and problems - steady state conduction through cylindrical wall and spherical wall derivation – problems - economic thickness of insulation – commonly used insulating materials. theory of convection – film concept of heat transfer temperature gradient in forced convection – derivation of overall heat transfer coefficient from individual heat transfer coefficient – simple problems. compute overall heat transfer coefficient – simple problems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interpret overall heat transfer coefficient. 2 Applying Heat transfer – Modes of heat transfer, definitions. Heat transfer by conduction in solids – Steady state and unsteady state flow – definition - Units of heat flow - Fourier’s law of conduction – Rate equation for heat flow – Steady state heat flow - Conduction through single wall – Derivation of equation and simple problems – Thermal conductivity – Units - Steady state conduction through composite wall in series - Derivation of equation and problems - Steady state conduction through cylindrical wall and spherical wall derivation – Problems - Economic thickness of insulation – Commonly used insulating materials. Theory of convection – Film concept of heat transfer temperature gradient in forced convection – Derivation of overall heat transfer coefficient from individual heat transfer coefficient – Simple problems. Compute overall heat transfer coefficient – Simple problems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Interpret overall heat transfer coefficient. 2 Applying Heat transfer – Modes of heat transfer, definitions. Heat transfer by conduction in solids – Steady state and unsteady state flow – definition - Units of heat flow - Fourier’s law of conduction – Rate equation for heat flow – Steady state heat flow - Conduction through single wall – Derivation of equation and simple problems – Thermal conductivity – Units - Steady state conduction through composite wall in series - Derivation of equation and problems - Steady state conduction through cylindrical wall and spherical wall derivation – Problems - Economic thickness of insulation – Commonly used insulating materials. Theory of convection – Film concept of heat transfer temperature gradient in forced convection – Derivation of overall heat transfer coefficient from individual heat transfer coefficient – Simple problems. Compute overall heat transfer coefficient – Simple problems. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
convection.

  M1.01     Outline the basic modes of heat transfer.                1
Understanding

            Interpret the mechanism of heat transfer through
  M1.02                                                              6
Applying
            conduction.
            Interpret the mechanism of heat transfer through
  M1.03                                                              3
Applying
            convection
  M1.04     Interpret overall heat transfer coefficient.             2
Applying

Contents:
Heat transfer – Modes of heat transfer, definitions.
Heat transfer by conduction in solids – Steady state and unsteady state flow –
definition -
Units of heat flow - Fourier’s law of conduction – Rate equation for heat flow –
Steady state
heat flow - Conduction through single wall – Derivation of equation and simple
problems –
Thermal conductivity – Units - Steady state conduction through composite wall in
series -
Derivation of equation and problems - Steady state conduction through cylindrical
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Summarise the heat transfer in fluids and radiation heat transfer.

This module is presented from the official syllabus scope for Process Heat Transfer. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise the heat transfer in fluids and radiation heat transfer.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

6 Applying and forced convection. Outline the mechanism of heat transfer in boiling

6 Applying and forced convection. Outline the mechanism of heat transfer in boiling is an official topic in Module 2 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying and forced convection. Outline the mechanism of heat transfer in boiling using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying and forced convection. outline the mechanism of heat transfer in boiling should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying and forced convection. Outline the mechanism of heat transfer in boiling means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 6 Applying and forced convection. Outline the mechanism of heat transfer in boiling □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding liquid.

4 Understanding liquid. is an official topic in Module 2 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding liquid. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding liquid. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding liquid. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Understanding liquid. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline condensation.

Outline condensation. is an official topic in Module 2 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline condensation. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline condensation. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline condensation. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Outline condensation. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Interpret radiation heat transfer. 5 Applying Mechanism of natural convection – Mechanism of Forced convection - Heat transfer by forced convection inside tubes – Heat transfer to fluid without phase change – Dimensionless equation-Reynolds number – Nusselt number -Prandtl number- Grashof number - Film coefficients in pipes laminar flow - sieder –Tate equation Film coefficient in turbulent flow – Dittus Boelters Equation – Simple problems. Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated boiling – Regimes of boiling. Condensation – Types of condensation Radiation – Elementary idea of black body – Gray body - Emissivity – Emissive power – Laws of radiation – Kirchoffs law, Planks law, Stefan Boltzmann law, Weins displacement law – Simple problems – Combined heat losses by convection and radiation

Interpret radiation heat transfer. 5 Applying Mechanism of natural convection – Mechanism of Forced convection - Heat transfer by forced convection inside tubes – Heat transfer to fluid without phase change – Dimensionless equation-Reynolds number – Nusselt number -Prandtl number- Grashof number - Film coefficients in pipes laminar flow - sieder –Tate equation Film coefficient in turbulent flow – Dittus Boelters Equation – Simple problems. Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated boiling – Regimes of boiling. Condensation – Types of condensation Radiation – Elementary idea of black body – Gray body - Emissivity – Emissive power – Laws of radiation – Kirchoffs law, Planks law, Stefan Boltzmann law, Weins displacement law – Simple problems – Combined heat losses by convection and radiation is an official topic in Module 2 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interpret radiation heat transfer. 5 Applying Mechanism of natural convection – Mechanism of Forced convection - Heat transfer by forced convection inside tubes – Heat transfer to fluid without phase change – Dimensionless equation-Reynolds number – Nusselt number -Prandtl number-Grashof number - Film coefficients in pipes laminar flow - sieder –Tate equation Film coefficient in turbulent flow – Dittus Boelters Equation – Simple problems.

Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated boiling – Regimes of boiling. Condensation – Types of condensation Radiation – Elementary idea of black body – Gray body - Emissivity – Emissive power – Laws of radiation – Kirchoffs law, Planks law, Stefan Boltzmann law, Weins displacement law – Simple problems – Combined heat losses by convection and radiation using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interpret radiation heat transfer. 5 applying mechanism of natural convection – mechanism of forced convection - heat transfer by forced convection inside tubes – heat transfer to fluid without phase change – dimensionless equation-reynolds number – nusselt number -prandtl number- grashof number - film coefficients in pipes laminar flow - sieder –tate equation film coefficient in turbulent flow – dittus boelters equation – simple problems. heat transfer in boiling liquids – flash boiling - sub cooled boiling – saturated boiling – regimes of boiling. condensation – types of condensation radiation – elementary idea of black body – gray body - emissivity – emissive power – laws of radiation – kirchoffs law, planks law, stefan boltzmann law, weins displacement law – simple problems – combined heat losses by convection and radiation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interpret radiation heat transfer. 5 Applying Mechanism of natural convection – Mechanism of Forced convection - Heat transfer by forced convection inside tubes – Heat transfer to fluid without phase change – Dimensionless equation-Reynolds number – Nusselt number -Prandtl number- Grashof number - Film coefficients in pipes laminar flow - sieder –Tate equation Film coefficient in turbulent flow – Dittus Boelters Equation – Simple problems. Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated boiling – Regimes of boiling. Condensation – Types of condensation Radiation – Elementary idea of black body – Gray body - Emissivity – Emissive power – Laws of radiation – Kirchoffs law, Planks law, Stefan Boltzmann law, Weins displacement law – Simple problems – Combined heat losses by convection and radiation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Interpret radiation heat transfer. 5 Applying Mechanism of natural convection – Mechanism of Forced convection - Heat transfer by forced convection inside tubes – Heat transfer to fluid without phase change – Dimensionless equation-Reynolds number – Nusselt number -Prandtl number- Grashof number - Film coefficients in pipes laminar flow - sieder –Tate equation Film coefficient in turbulent flow – Dittus Boelters Equation – Simple problems. Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated boiling – Regimes of boiling. Condensation – Types of condensation Radiation – Elementary idea of black body – Gray body - Emissivity – Emissive power – Laws of radiation – Kirchoffs law, Planks law, Stefan Boltzmann law, Weins displacement law – Simple problems – Combined heat losses by convection and radiation □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarise the heat transfer in fluids and radiation heat transfer.
            Interpret the mechanism of heat transfer by natural
  M2.01                                                               6
Applying
            and forced convection.
            Outline the mechanism of heat transfer in boiling
  M2.02                                                               4
Understanding
            liquid.
  M2.03     Outline condensation.                                   2
Understanding

  M2.04     Interpret radiation heat transfer.                      5
Applying
            Series Test – I

Contents:
Mechanism of natural convection – Mechanism of Forced convection - Heat transfer
by
forced convection inside tubes – Heat transfer to fluid without phase change –
Dimensionless equation-Reynolds number – Nusselt number -Prandtl number- Grashof
number - Film coefficients in pipes laminar flow - sieder –Tate equation Film
coefficient in
turbulent flow – Dittus Boelters Equation – Simple problems.
Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated
boiling –
Regimes of boiling.
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarise heat exchangers.

This module is presented from the official syllabus scope for Process Heat Transfer. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise heat exchangers.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Understand the use of heat exchangers in industry. 1 Understanding Illustrate major types of heat exchangers in

Understand the use of heat exchangers in industry. 1 Understanding Illustrate major types of heat exchangers in is an official topic in Module 3 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the use of heat exchangers in industry. 1 Understanding Illustrate major types of heat exchangers in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the use of heat exchangers in industry. 1 understanding illustrate major types of heat exchangers in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand the use of heat exchangers in industry. 1 Understanding Illustrate major types of heat exchangers in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Understand the use of heat exchangers in industry. 1 Understanding Illustrate major types of heat exchangers in [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

6 Understanding industry. Identify the flow patterns and allocation of fluids

6 Understanding industry. Identify the flow patterns and allocation of fluids is an official topic in Module 3 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding industry. Identify the flow patterns and allocation of fluids using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding industry. identify the flow patterns and allocation of fluids should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding industry. Identify the flow patterns and allocation of fluids means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 6 Understanding industry. Identify the flow patterns and allocation of fluids □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding in heat exchangers.

3 Understanding in heat exchangers. is an official topic in Module 3 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding in heat exchangers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding in heat exchangers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding in heat exchangers. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding in heat exchangers. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Compute heat transfer area for heat exchangers. 6 Applying Heat exchangers – Uses of heat exchangers in industry Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube heat exchanger - Floating head heat exchanger - Plate type heat exchanger and finned tube heat exchanger– Identify the functions of tube sheet, tie road and baffles in heat exchangers – Identify the types of baffles. Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross flow heat exchangers – Single pass and multi pass heat exchangers – Naming of heat exchanger based on number of passes– Illustrate flow pattern in multi pass heat exchangers- Outline the routing of shell and tube side fluids in heat exchangers. Temperature distribution in heat exchangers, LMTD - Estimation of heat transfer area and length of tubes for heat exchangers (LMTD method) – Effect of fouling in heat exchangers.

Compute heat transfer area for heat exchangers. 6 Applying Heat exchangers – Uses of heat exchangers in industry Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube heat exchanger - Floating head heat exchanger - Plate type heat exchanger and finned tube heat exchanger– Identify the functions of tube sheet, tie road and baffles in heat exchangers – Identify the types of baffles. Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross flow heat exchangers – Single pass and multi pass heat exchangers – Naming of heat exchanger based on number of passes– Illustrate flow pattern in multi pass heat exchangers- Outline the routing of shell and tube side fluids in heat exchangers. Temperature distribution in heat exchangers, LMTD - Estimation of heat transfer area and length of tubes for heat exchangers (LMTD method) – Effect of fouling in heat exchangers. is an official topic in Module 3 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compute heat transfer area for heat exchangers. 6 Applying Heat exchangers – Uses of heat exchangers in industry Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube heat exchanger - Floating head heat exchanger - Plate type heat exchanger and finned tube heat exchanger– Identify the functions of tube sheet, tie road and baffles in heat exchangers –

Identify the types of baffles. Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross flow heat exchangers – Single pass and multi pass heat exchangers – Naming of heat exchanger based on number of passes– Illustrate flow pattern in multi pass heat exchangers- Outline the routing of shell and tube side fluids in heat exchangers. Temperature distribution in heat exchangers, LMTD - Estimation of heat transfer area and length of tubes for heat exchangers (LMTD method) – Effect of fouling in heat exchangers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compute heat transfer area for heat exchangers. 6 applying heat exchangers – uses of heat exchangers in industry illustrate double pipe heat exchanger - shell and tube heat exchanger - u tube heat exchanger - floating head heat exchanger - plate type heat exchanger and finned tube heat exchanger– identify the functions of tube sheet, tie road and baffles in heat exchangers – identify the types of baffles. flow patterns in heat exchangers - parallel flow – counter current flow – cross flow heat exchangers – single pass and multi pass heat exchangers – naming of heat exchanger based on number of passes– illustrate flow pattern in multi pass heat exchangers- outline the routing of shell and tube side fluids in heat exchangers. temperature distribution in heat exchangers, lmtd - estimation of heat transfer area and length of tubes for heat exchangers (lmtd method) – effect of fouling in heat exchangers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compute heat transfer area for heat exchangers. 6 Applying Heat exchangers – Uses of heat exchangers in industry Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube heat exchanger - Floating head heat exchanger - Plate type heat exchanger and finned tube heat exchanger– Identify the functions of tube sheet, tie road and baffles in heat exchangers – Identify the types of baffles. Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross flow heat exchangers – Single pass and multi pass heat exchangers – Naming of heat exchanger based on number of passes– Illustrate flow pattern in multi pass heat exchangers- Outline the routing of shell and tube side fluids in heat exchangers. Temperature distribution in heat exchangers, LMTD - Estimation of heat transfer area and length of tubes for heat exchangers (LMTD method) – Effect of fouling in heat exchangers. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Compute heat transfer area for heat exchangers. 6 Applying Heat exchangers – Uses of heat exchangers in industry Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube heat exchanger - Floating head heat exchanger - Plate type heat exchanger and finned tube heat exchanger– Identify the functions of tube sheet, tie road and baffles in heat exchangers – Identify the types of baffles. Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross flow heat exchangers – Single pass and multi pass heat exchangers – Naming of heat exchanger based on number of passes– Illustrate flow pattern in multi pass heat exchangers- Outline the routing of shell and tube side fluids in heat exchangers. Temperature distribution in heat exchangers, LMTD - Estimation of heat transfer area and length of tubes for heat exchangers (LMTD method) – Effect of fouling in heat exchangers. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarise heat exchangers.

  M3.01     Understand the use of heat exchangers in industry.     1
Understanding
            Illustrate major types of heat exchangers in
  M3.02                                                            6
Understanding
            industry.
            Identify the flow patterns and allocation of fluids
  M3.03                                                            3
Understanding
            in heat exchangers.

  M3.04     Compute heat transfer area for heat exchangers.        6
Applying

Contents:
Heat exchangers – Uses of heat exchangers in industry
Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube
heat
exchanger - Floating head heat exchanger - Plate type heat exchanger and finned
tube heat
exchanger– Identify the functions of tube sheet, tie road and baffles in heat
exchangers –
Identify the types of baffles.
Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross
flow heat
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Summarise evaporators.

This module is presented from the official syllabus scope for Process Heat Transfer. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise evaporators.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Understand the use of evaporators in industry.

Understand the use of evaporators in industry. is an official topic in Module 4 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the use of evaporators in industry. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the use of evaporators in industry. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand the use of evaporators in industry. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Understand the use of evaporators in industry. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate different types of evaporators.

Illustrate different types of evaporators. is an official topic in Module 4 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate different types of evaporators. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate different types of evaporators. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate different types of evaporators. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Illustrate different types of evaporators. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Compute heat transfer area for an evaporator.

Compute heat transfer area for an evaporator. is an official topic in Module 4 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compute heat transfer area for an evaporator. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compute heat transfer area for an evaporator. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compute heat transfer area for an evaporator. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Compute heat transfer area for an evaporator. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Illustrate multiple effect evaporators. 4 Understanding Evaporators – Uses of evaporators in industry. Constructional Details and Working Of Evaporator - Types of evaporators– Basis of classification – Horizontal tube – Vertical tube – Climbing film – Falling film – Forced circulation evaporators – Examples of application of each evaporator in industries. Material and heat balance equation for single effect evaporators – Simple problems – Calculation of heating area – Define: capacity and steam economy. Multiple effect evaporation – Principle of multiple effect evaporation – Methods of feeding – Advantages of multiple effect evaporation - Temperature distribution in multiple effect and effect of boiling point elevation. Text / Reference T/R Book Title/Author T1 Heat Transfer; K A Ghavane R1 Heat transfer and its application; Binay K Dutta R2 Introduction to Chemical Engineering; Water.L.Badger R3 Unit operation of Chemical Engineering; Warren.L.McCabe andSmith R4 Heat transfer; D.D.Tiwari Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/103/103103032/

Illustrate multiple effect evaporators. 4 Understanding Evaporators – Uses of evaporators in industry. Constructional Details and Working Of Evaporator - Types of evaporators– Basis of classification – Horizontal tube – Vertical tube – Climbing film – Falling film – Forced circulation evaporators – Examples of application of each evaporator in industries. Material and heat balance equation for single effect evaporators – Simple problems – Calculation of heating area – Define: capacity and steam economy. Multiple effect evaporation – Principle of multiple effect evaporation – Methods of feeding – Advantages of multiple effect evaporation - Temperature distribution in multiple effect and effect of boiling point elevation. Text / Reference T/R Book Title/Author T1 Heat Transfer; K A Ghavane R1 Heat transfer and its application; Binay K Dutta R2 Introduction to Chemical Engineering; Water.L.Badger R3 Unit operation of Chemical Engineering; Warren.L.McCabe andSmith R4 Heat transfer; D.D.Tiwari Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/103/103103032/ is an official topic in Module 4 of Process Heat Transfer. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate multiple effect evaporators. 4 Understanding Evaporators – Uses of evaporators in industry. Constructional Details and Working Of Evaporator - Types of evaporators– Basis of classification – Horizontal tube – Vertical tube – Climbing film – Falling film – Forced circulation evaporators – Examples of application of each evaporator in industries. Material and heat balance equation for single effect evaporators – Simple problems – Calculation of heating area – Define: capacity and steam economy. Multiple effect evaporation – Principle of multiple effect evaporation – Methods of feeding – Advantages of multiple effect evaporation - Temperature distribution in multiple effect and effect of boiling point elevation. Text / Reference T/R Book Title/Author T1 Heat Transfer; K A Ghavane R1 Heat transfer and its application; Binay K Dutta R2 Introduction to Chemical Engineering; Water.L.Badger R3 Unit operation of Chemical Engineering; Warren.L.McCabe andSmith R4 Heat transfer; D.D.Tiwari Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/103/103103032/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate multiple effect evaporators. 4 understanding evaporators – uses of evaporators in industry. constructional details and working of evaporator - types of evaporators– basis of classification – horizontal tube – vertical tube – climbing film – falling film – forced circulation evaporators – examples of application of each evaporator in industries. material and heat balance equation for single effect evaporators – simple problems – calculation of heating area – define: capacity and steam economy. multiple effect evaporation – principle of multiple effect evaporation – methods of feeding – advantages of multiple effect evaporation - temperature distribution in multiple effect and effect of boiling point elevation. text / reference t/r book title/author t1 heat transfer; k a ghavane r1 heat transfer and its application; binay k dutta r2 introduction to chemical engineering; water.l.badger r3 unit operation of chemical engineering; warren.l.mccabe andsmith r4 heat transfer; d.d.tiwari online resources sl.no website link 1 https://nptel.ac.in/courses/103/103/103103032/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate multiple effect evaporators. 4 Understanding Evaporators – Uses of evaporators in industry. Constructional Details and Working Of Evaporator - Types of evaporators– Basis of classification – Horizontal tube – Vertical tube – Climbing film – Falling film – Forced circulation

Aspect	What to prepare
	evaporators – Examples of application of each evaporator in industries. Material and heat balance equation for single effect evaporators – Simple problems – Calculation of heating area – Define: capacity and steam economy. Multiple effect evaporation – Principle of multiple effect evaporation – Methods of feeding – Advantages of multiple effect evaporation - Temperature distribution in multiple effect and effect of boiling point elevation. Text / Reference T/R Book Title/Author T1 Heat Transfer; K A Ghavane R1 Heat transfer and its application; Binay K Dutta R2 Introduction to Chemical Engineering; Water.L.Badger R3 Unit operation of Chemical Engineering; Warren.L.McCabe andSmith R4 Heat transfer; D.D.Tiwari Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/103/103103032/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate multiple effect evaporators. 4 Understanding Evaporators – Uses of evaporators in industry. Constructional Details and Working Of Evaporator - Types of evaporators– Basis of classification – Horizontal tube – Vertical tube – Climbing film – Falling film – Forced circulation evaporators – Examples of application of each evaporator in industries. Material and heat balance equation for single effect evaporators – Simple problems – Calculation of heating area – Define: capacity and steam economy. Multiple effect evaporation – Principle of multiple effect evaporation – Methods of feeding – Advantages of multiple effect evaporation - Temperature distribution in multiple effect and effect of boiling point elevation. Text / Reference T/R Book Title/Author T1 Heat Transfer; K A Ghavane R1 Heat transfer and its application; Binay K Dutta R2 Introduction to Chemical Engineering; Water.L.Badger R3 Unit operation of Chemical Engineering; Warren.L.McCabe andSmith R4 Heat transfer; D.D.Tiwari Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/103/103103032/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarise evaporators.

  M 4.01    Understand the use of evaporators in industry.        1
Understanding

  M 4.02    Illustrate different types of evaporators.            6
Understanding

  M 4.03    Compute heat transfer area for an evaporator.         4
Applying

  M 4.04    Illustrate multiple effect evaporators.               4
Understanding

            Series Test – II

Contents:
Evaporators – Uses of evaporators in industry.
Constructional Details and Working Of Evaporator - Types of evaporators– Basis of
classification – Horizontal tube – Vertical tube – Climbing film – Falling film –
Forced
circulation evaporators – Examples of application of each evaporator in
industries.
Material and heat balance equation for single effect evaporators – Simple
problems –
Calculation of heating area – Define: capacity and steam economy.
Multiple effect evaporation – Principle of multiple effect evaporation – Methods
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Outline the basic modes of heat transfer. 1 Understanding Interpret the mechanism of heat transfer through. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the basic modes of heat transfer. 1 Understanding Interpret the mechanism of heat transfer through*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Applying conduction. Interpret the mechanism of heat transfer through. (3 marks)

Answer: Begin with the standard definition or identity of *6 Applying conduction. Interpret the mechanism of heat transfer through*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying convection. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying convection*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Interpret overall heat transfer coefficient. 2 Applying Heat transfer – Modes of heat transfer, definitions. Heat transfer by conduction in solids – Steady state and unsteady state flow – definition - Units of heat flow - Fourier’s law of conduction – Rate equation for heat flow – Steady state heat flow - Conduction through single wall – Derivation of equation and simple problems – Thermal conductivity – Units - Steady state conduction through composite wall in series - Derivation of equation and problems - Steady state conduction through cylindrical wall and spherical wall derivation – Problems - Economic thickness of insulation – Commonly used insulating materials. Theory of convection – Film concept of heat transfer temperature gradient in forced convection – Derivation of overall heat transfer coefficient from individual heat transfer coefficient – Simple problems. Compute overall heat transfer coefficient – Simple problems.. (3 marks)

Answer: Begin with the standard definition or identity of *Interpret overall heat transfer coefficient. 2 Applying Heat transfer – Modes of heat transfer, definitions. Heat transfer by conduction in solids – Steady state and unsteady state flow – definition - Units of heat flow - Fourier's law of conduction – Rate equation for heat flow – Steady state heat flow - Conduction through single wall – Derivation of equation and simple problems – Thermal conductivity – Units - Steady state conduction through composite wall in series - Derivation of equation and problems - Steady state conduction through cylindrical wall and spherical wall derivation – Problems - Economic thickness of insulation – Commonly used insulating materials. Theory of convection – Film concept of heat transfer temperature gradient in forced convection – Derivation of overall heat transfer coefficient from individual heat transfer coefficient – Simple problems. Compute overall heat transfer coefficient – Simple problems.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 6 Applying and forced convection. Outline the mechanism of heat transfer in boiling. (3 marks)

Answer: Begin with the standard definition or identity of *6 Applying and forced convection. Outline the mechanism of heat transfer in boiling*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding liquid.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding liquid.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline condensation.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline condensation.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Interpret radiation heat transfer. 5 Applying Mechanism of natural convection – Mechanism of Forced convection - Heat transfer by forced convection inside tubes – Heat transfer to fluid without phase change – Dimensionless equation-Reynolds number – Nusselt number -Prandtl number-Grashof number - Film coefficients in pipes laminar flow - sieder –Tate equation Film coefficient in turbulent flow – Dittus Boelters Equation – Simple problems. Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated boiling – Regimes of boiling. Condensation – Types of condensation Radiation – Elementary idea of black body – Gray body - Emissivity – Emissive power – Laws of radiation – Kirchoffs law, Planks law, Stefan Boltzmann law, Weins displacement law – Simple problems – Combined heat losses by convection and radiation. (3 marks)

Answer: Begin with the standard definition or identity of *Interpret radiation heat transfer. 5 Applying Mechanism of natural convection – Mechanism of Forced convection - Heat transfer by forced convection inside tubes – Heat transfer to fluid without phase change – Dimensionless equation-Reynolds number – Nusselt number -Prandtl number-Grashof number - Film coefficients in pipes laminar flow - sieder –Tate equation Film coefficient in turbulent flow – Dittus Boelters Equation – Simple problems. Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated boiling – Regimes of boiling. Condensation – Types of condensation Radiation – Elementary idea of black body – Gray body - Emissivity – Emissive power – Laws of radiation – Kirchoffs law, Planks law, Stefan Boltzmann law, Weins displacement law – Simple problems – Combined heat losses by convection and radiation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Understand the use of heat exchangers in industry. 1 Understanding Illustrate major types of heat exchangers in. (3 marks)

Answer: Begin with the standard definition or identity of *Understand the use of heat exchangers in industry. 1 Understanding Illustrate major types of heat exchangers in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding industry. Identify the flow patterns and allocation of fluids. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding industry. Identify the flow patterns and allocation of fluids*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding in heat exchangers.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding in heat exchangers.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Compute heat transfer area for heat exchangers. 6 Applying Heat exchangers – Uses of heat exchangers in industry Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube heat exchanger - Floating head heat exchanger - Plate type heat exchanger and finned tube heat exchanger– Identify the functions of tube sheet, tie road and baffles in heat exchangers – Identify the types of baffles. Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross flow heat exchangers – Single pass and multi pass heat exchangers – Naming of heat exchanger based on number of passes– Illustrate flow pattern in multi pass heat exchangers- Outline the routing of shell and tube side fluids in heat exchangers. Temperature distribution in heat exchangers, LMTD - Estimation of heat transfer area and length of tubes for heat exchangers (LMTD method) – Effect of fouling in heat exchangers.. (3 marks)

Answer: Begin with the standard definition or identity of *Compute heat transfer area for heat exchangers. 6 Applying Heat exchangers – Uses of heat exchangers in industry Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube heat exchanger - Floating head heat exchanger - Plate type heat exchanger and finned tube heat exchanger– Identify the functions of tube sheet, tie road and baffles in heat exchangers – Identify the types of baffles. Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross flow heat exchangers – Single pass and multi pass heat exchangers – Naming of heat exchanger based on number of passes– Illustrate flow pattern in multi pass heat exchangers- Outline the routing of shell and tube side fluids in heat exchangers. Temperature distribution in heat exchangers, LMTD - Estimation of heat transfer area and length of tubes for heat exchangers (LMTD method) – Effect of fouling in heat exchangers.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Understand the use of evaporators in industry.. (3 marks)

Answer: Begin with the standard definition or identity of *Understand the use of evaporators in industry.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Illustrate different types of evaporators.. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate different types of evaporators.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Compute heat transfer area for an evaporator.. (3 marks)

Answer: Begin with the standard definition or identity of *Compute heat transfer area for an evaporator.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Illustrate multiple effect evaporators. 4 Understanding Evaporators – Uses of evaporators in industry. Constructional Details and Working Of Evaporator - Types of evaporators– Basis of classification – Horizontal tube – Vertical tube – Climbing film – Falling film – Forced circulation evaporators – Examples of application of each evaporator in industries. Material and heat balance equation for single effect evaporators – Simple problems – Calculation of heating area – Define: capacity and steam economy. Multiple effect evaporation – Principle of multiple effect evaporation – Methods of feeding – Advantages of multiple effect evaporation - Temperature

distribution in multiple effect and effect of boiling point elevation. Text / Reference T/R Book Title/Author T1 Heat Transfer; K A Ghavane R1 Heat transfer and its application; Binay K Dutta R2 Introduction to Chemical Engineering; Water.L.Badger R3 Unit operation of Chemical Engineering; Warren.L.McCabe andSmith R4 Heat transfer; D.D.Tiwari Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/103/103103032/. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate multiple effect evaporators. 4 Understanding Evaporators – Uses of evaporators in industry. Constructional Details and Working Of Evaporator - Types of evaporators– Basis of classification – Horizontal tube – Vertical tube – Climbing film – Falling film – Forced circulation evaporators – Examples of application of each evaporator in industries. Material and heat balance equation for single effect evaporators – Simple problems – Calculation of heating area – Define: capacity and steam economy. Multiple effect evaporation – Principle of multiple effect evaporation – Methods of feeding – Advantages of multiple effect evaporation - Temperature distribution in multiple effect and effect of boiling point elevation. Text / Reference T/R Book Title/Author T1 Heat Transfer; K A Ghavane R1 Heat transfer and its application; Binay K Dutta R2 Introduction to Chemical Engineering; Water.L.Badger R3 Unit operation of Chemical Engineering; Warren.L.McCabe andSmith R4 Heat transfer; D.D.Tiwari Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/103/103103032/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Outline the basic modes of heat transfer. 1 Understanding Interpret the mechanism of heat transfer through**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **6 Applying and forced convection. Outline the mechanism of heat transfer in boiling**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Understand the use of heat exchangers in industry. 1 Understanding Illustrate major types of heat exchangers in**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Understand the use of evaporators in industry.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://nptel.ac.in/courses/103/103/103103032/

IN-PAGE SEARCH

Search this lesson